

Chiyoung Kim

chiyoungk@andrew.cmu.edu ♦ (412) 475-6331 ♦ chiyoungk.github.io ♦ linkedin.com/in/chiyoungk

EDUCATION

Carnegie Mellon University

Visiting Scholar, Software and Societal Systems Department (S3D) Pittsburgh, PA
Aug 2026 – Present
Fully funded by an IITP–Sogang University research visit program (25 graduate students selected nationwide)

Chung-Ang University

M.S. in Intelligent Semiconductor Engineering (Expected Feb 2028) Seoul, South Korea
Mar 2026 – Present
GPA: 4.33/4.5
Selected Coursework: Intelligent Video Processing, Advanced Natural Language Processing

Tecnologico de Monterrey

Exchange Student Monterrey, Mexico
Aug 2023 – Jun 2024

Kookmin University

B.S. in Electrical Engineering Seoul, South Korea
Mar 2019 – Feb 2026
GPA: 3.85/4.5 (Dean's List for the final semester)
Mandatory military service, Republic of Korea Apr 2021 – Jan 2023

RESEARCH

Chung-Ang University

Graduate Research Assistant, GAIL (Advisor: Prof. Minhyeok Lee) Seoul, South Korea
Mar 2026 – Present

- Developed an offline recovery method for pruned VLA models via projector-free hidden-state distillation, restoring a 63%-pruned OpenVLA-OFT from 0.8% to 89.7% on LIBERO-Long in ~8 GPU-hours.
- Ran a nine-ratio compression sweep on two VLA backbones and a matched width-vs-depth pruning comparison, identifying 72% reduction as the deployment operating point.
- Collected 450 teleoperated demonstrations on an AgileX PiPER arm and deployed the compressed policy on its Jetson AGX Thor: 77.5% success, 2.23× lower latency, 62% less memory than the dense model.
- Co-led ReShoot, a video-diffusion pipeline that re-renders recorded demonstrations under new appearances, raising recolored-object success on two real robots from 0% to 42.9% and 47.5%.
- Contributed to CoTinyVLA, a 0.9B-parameter VLA trained with chain-of-thought distillation that outperforms 7B baselines on all four LIBERO-Plus suites.

PUBLICATIONS

Chiyoung Kim, Sanghyuk Roy Choi, and Minhyeok Lee. "Recovering Aggressively Pruned Vision-Language-Action Models with Offline Hidden-State Distillation." *arXiv preprint arXiv:2609.19579*, 2026. (Under review, ICRA 2027)

Chiyoung Kim*, Min Sung Choi*, Jinho Ju, Chanhoe Gu, Donghwan Hwang, Wonseok Choi, Woongsun Jeon, and Minhyeok Lee. "ReShoot: Generative Visual Domain Randomization of Recorded Robot Demonstrations for Visuomotor Policy Learning." *arXiv preprint arXiv:2609.19661*, 2026. (Under review, ICRA 2027) *Equal contribution.

Minhyeok Lee, **Chiyoung Kim**, Chanhoe Gu, Seongrok Kim, Sanghyuk Roy Choi, Donghwan Hwang, Donghun Ryu, and Seokhyun Kim. "CoTinyVLA: Chain-of-Thought Distillation for a Sub-Billion-Parameter Vision-Language-Action Model." *arXiv preprint arXiv:2607.25487*, 2026.

SKILLS

Programming & Tools: Python, PyTorch, Linux

Vision-Language-Action Models: OpenVLA, OpenVLA-OFT, CogACT, imitation learning, LoRA fine-tuning, structured pruning, hidden-state distillation

Robot Systems & Deployment: AgileX PiPER manipulator, teleoperation, demonstration data collection, NVIDIA Jetson AGX Thor, Raspberry Pi 5, on-device inference

Simulation & Benchmarking: LIBERO, LIBERO-Plus, SimplerEnv, MuJoCo

Languages: Korean (Native Speaker), English (Fluent), Spanish (Intermediate)

TEACHING ASSISTANT EXPERIENCE

Chung-Ang University	Seoul, South Korea
Teaching Assistant – Embedded Systems Design Lab (Raspberry Pi 5)	Mar 2026 – Jun 2026
• Led weekly hands-on Raspberry Pi 5 lab sessions for 23 undergraduate students, guiding hardware setup, embedded programming, and debugging. • Maintained the course lab manual and authored and graded the final examination and term-project evaluation for all students.	

LEADERSHIP

President , Table Tennis Club (Student Club) – Kookmin University	Sep 2019 – Dec 2019
• Led the university's table tennis club, organizing twice-weekly training sessions and team gatherings for members throughout the semester.	

AWARDS AND HONORS

Research Visit Fellowship, IITP and Sogang University	2026
Full funding for a semester-long research visit at Carnegie Mellon University; 25 graduate students selected nationwide (Institute for Information & Communications Technology Planning & Evaluation, Republic of Korea)	
Dean's List, Kookmin University	Fall 2025
English Proficiency Scholarship (4 semesters), Kookmin University	2020 – 2025
Faculty Assistantship Scholarship, Kookmin University	Fall 2024
Table Tennis , Lower-Div. Doubles: 1st Place (2019), Runner-up (2020), 8-University Tournament; Runner-up, National Collegiate Tournament (2025)	